ProteinGuide

Property guidance for protein sequence generative models

Xiong, Gaur, Lukarska, Nisonoff et al. — *Nature Biotechnology*, 2026

Journal Club DOI: 10.1038/s41587-026-03207-z

Central idea: steer a pretrained protein generator during sampling with a separate property predictor.

The engineering question

A protein generator knows what plausible proteins look like. An assay tells us which variants perform the task we care about.

1. How can we combine these two sources of information without retraining the generator?
2. Can this produce useful variants beyond the measured examples?
3. Can we impose two competing property constraints at once?

Contribution: a statistical framework connecting masked language, any-order autoregressive, masked diffusion and flow models, plus practical guidance algorithms.

Paper abstract; arXiv:2505.04823v4, introduction and Results.

Why changing the sampling process could help

Approach	What changes?	Main practical constraint
Post hoc filtering	Rank completed samples using a predictor	Rare desirable sequences may scarcely be sampled.
Fine-tuning	Continue training the generator on selected examples	Requires training and model selection; may concentrate around observed examples.
ProteinGuide	Reweight choices along the generation trajectory	Requires a predictor that works on partially generated sequences.

Fine-tuning remains a strong baseline for some tasks; these experiments do not establish a universal ranking.

Fig. 1c; preprint Results and Methods (common experimental setup).

Inputs, output and the conditional distribution

Inputs

A pretrained generator $p_\theta(x)$ for sequence x.

A predictor $p_\phi(y \mid x)$ for a desired outcome y (for example, stability above a threshold).

Optional structural context or fixed residues.

Output

A library biased toward the requested outcome:

$$p(x \mid y) \propto p_\theta(x) p_\phi(y \mid x).$$

The generator supplies a prior over plausible sequences; the predictor supplies task information.

The predictor is trained or supplied separately. "No retraining" refers to the *generator*, not to the entire workflow.

Fig. 1a; preprint "Intuitive overview of guidance".

A concrete sampling step: reweight, normalize, sample

Suppose the partial sequence is `M ? K S`; we will fill the second position.

Candidate residue	Generator	Probability of desired outcome	Product
A	0.60	0.10	0.06
D	0.30	0.70	0.21
V	0.10	0.30	0.03

Normalize the products by 0.30: the guided probabilities become $(0.20, 0.70, 0.10)$. D becomes likely because the *combination* of plausibility and desired-property probability favors it. Repeat using the updated context.

Illustrative numbers, not experimental data. Simplified discrete guidance update from the paper's framework.

The predictor must understand incomplete sequences

Let c be the already revealed residues and a a candidate at the next position i:

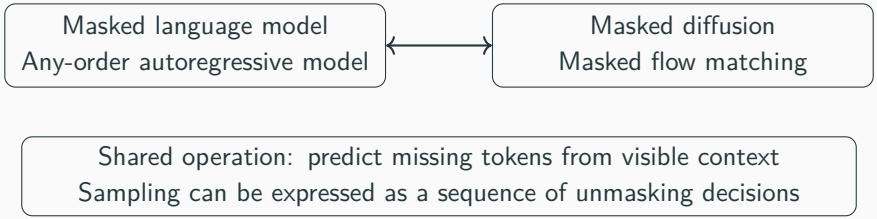
$$p(a \mid c, y) \propto p_\theta(a \mid c)\, p_\phi(y \mid c, a).$$

1. Train the property predictor with random masking so that it accepts partial sequences.
2. Obtain candidate amino-acid probabilities from the pretrained generator.
3. Evaluate how each candidate changes the desired-property likelihood.
4. Sample from the normalized product, then reveal another position.

An ordinary full-sequence fitness predictor cannot automatically supply calibrated probabilities at every masking level.

Preprint Methods, guidance and noisy predictor training; published SI, derivations.

What makes this usable across model classes?



The exact sampling result concerns the paper's masking process with minimal transitions: a residue is revealed once, without corrective remasking. It is not a claim that arbitrary noise processes or every default heuristic sampler are identical.

Fig. 1b; published SI, Section 1.2 (Proposition 1 and lemmas).

Two practical guidance algorithms

	Discrete-time exact guidance (DEG)	Taylor-approximate guidance (TAG)
Computation	Evaluate candidate changes explicitly	Approximate likelihood changes with one predictor gradient
Benefit	Avoids the first-order predictor approximation	Efficient when the alphabet or predictor is large
Tradeoff	More candidate evaluations	Approximation and numerical integration can matter

Guidance strength γ scales the property signal. Bayes conditioning corresponds to $\gamma = 1$ under the ideal model; stronger guidance deliberately changes the weighting.

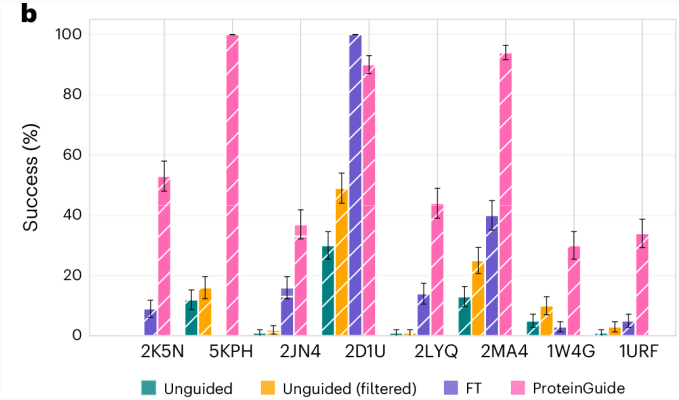
Preprint Results and Methods; published SI derivations and discussion of DEG versus TAG.

The evidence asks progressively harder questions

Setting	Design task	What is measured?
Interpolation	ProteinMPNN: improve stability on eight backbones	Oracle stability and predicted refolding
Extrapolation	ESM3: CreiLOV, PABP, UBE4B properties	Oracle fitness, novelty and fold confidence
Multiple properties	ESM C: PbrR lead/zinc response	Oracle target-region membership and diversity
Laboratory feedback	TadA adenine base editors	Bacterial selection, then individual validation

Figs. 2–5; Extended Data Table 1. The first three rows are computational evaluations.

Stability: improve the property while preserving the fold



Eight mini-protein backbones; $k = 100$ samples per method and backbone.

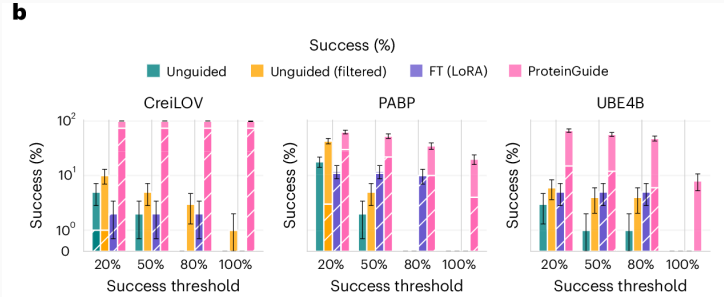
Bar height: predicted $\Delta\Delta G \leq 0$.

Hatching: also refolds within 2 Å of the target.

Guidance improves most targets; FT wins on 2D1U.

Published Fig. 2b, cropped with axes and legend. Errors: one standard error of the empirical success proportion.

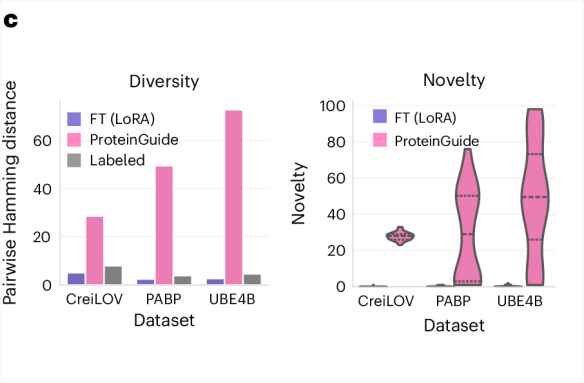
Extrapolation: oracle fitness beyond the labeled range



Success requires a unique, novel sequence above a specified percentile of the full labeled dataset. Hatching adds predicted pLDDT ≥ 0.8. Guidance retains successes at the strictest threshold across all three tasks. This demonstrates *oracle-predicted* extrapolation; new sequences were not assayed here.

Published Fig. 3b; percentiles and pLDDT definition from its caption. Log-scaled success axes.

Novelty matters, but needs to accompany function



Diversity: average pairwise Hamming distance among generated sequences.

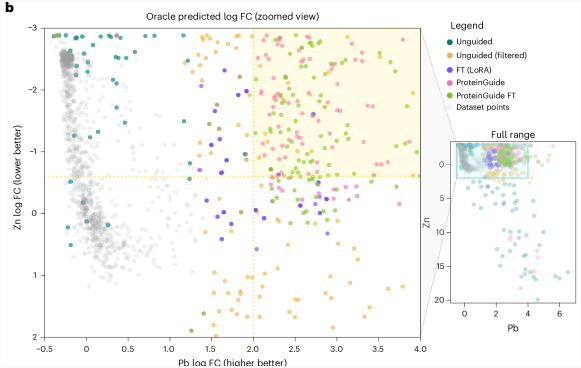
Novelty: minimum Hamming distance to any labeled sequence.

Guidance produces more diverse and more novel libraries than FT in these assays.

Large Hamming distance alone does not demonstrate retained biochemical function.

Published Fig. 3c. These distances count amino-acid differences; they are not functional measurements.

PbrR: increase lead response, decrease zinc response



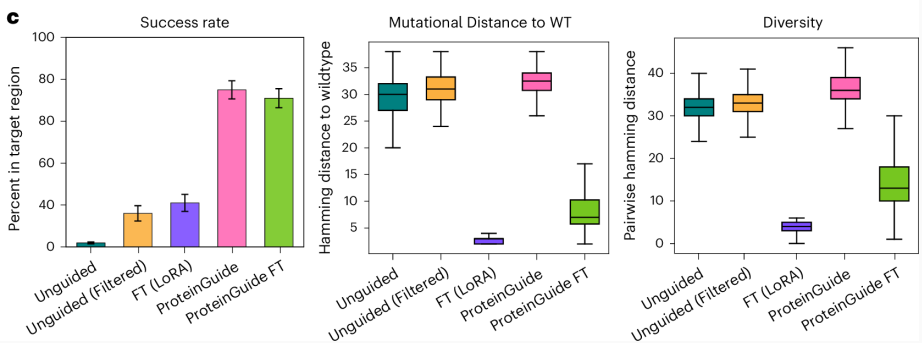
Desired region:
Pb log FC ≥ 2.0,
Zn log FC ≤ -0.6.

The zinc axis is reversed: upward is better.

Gray points: initial experimental library. Colored points: *oracle predictions* for designs.

Published Fig. 4b (crop). The journal caption specifies normalization with wild-type log FC = 1.0.

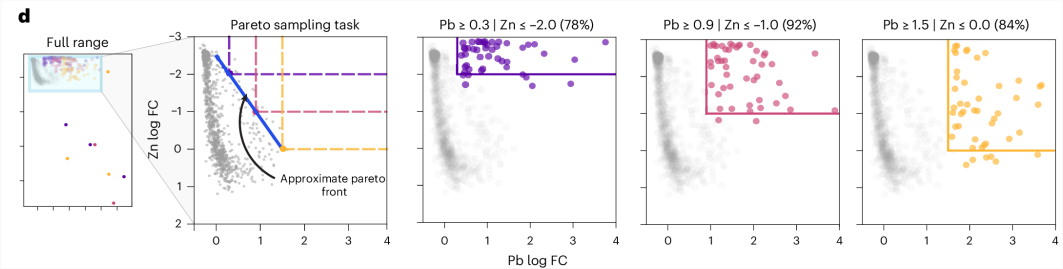
PbrR: success and diversity tell different stories



ProteinGuide has roughly three-quarters of samples in the target region and high diversity. Guidance after fine-tuning also succeeds, with less diversity. The published comparison includes successful FT designs: "fine-tuning cannot extrapolate" would overstate this evidence.

Published Fig. 4c, approximate bar reading; no preprint baseline percentages substituted.

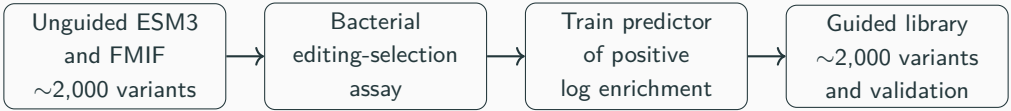
Change the objective without retraining the generator



Different target quadrants give reported oracle success rates of 78%, 92% and 84%. The same framework can be queried at different tradeoffs. This is conditional sampling around specified targets. It does not prove that the true biochemical Pareto frontier has been found.

Published Fig. 4d. Axes remain the paper's Pb/Zn response coordinates.

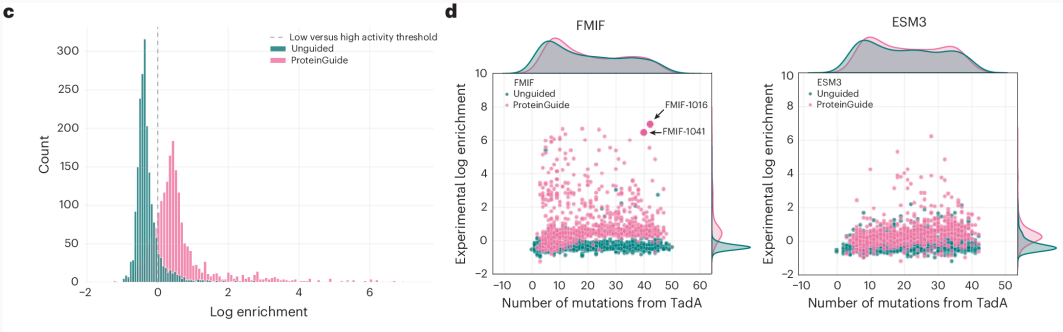
Wet-lab test: one feedback cycle for TadA



Design the 86 C-terminal residues of TadA while conditioning on its structure. FMIF training excluded engineered ABE examples to reduce prior exposure to known improvements. Editing restores antibiotic resistance; enrichment reflects survival under selection relative to a nonselective condition.

Fig. 5a; preprint Methods, Section 1.11. Two experimental libraries, one guidance update.

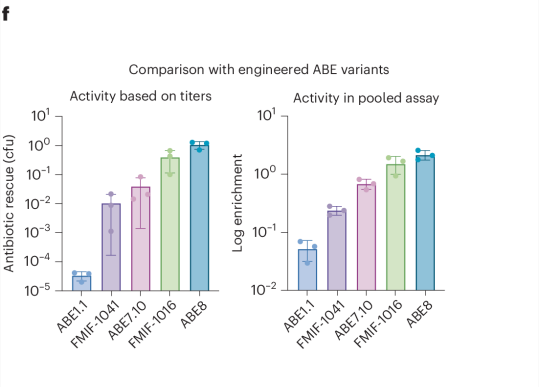
Guided libraries contain more active variants



The activity distribution shifts and FMIF yields high-enrichment outliers over a range of mutation counts. Predictions alone (Fig. 5b) would be insufficient; these panels show experimental enrichment.

Published Fig. 5c,d. Teal: unguided; pink: guided. Enrichment is an assay readout, not direct mammalian editing efficiency.

Individual validation supports a stronger claim



Published Fig. 5f; original labels and logarithmic axes retained.

Two FMIF variants were compared with ABE1.1, ABE7.10 and ABE8.

FMIF-1016 exceeds ABE7.10 and remains below ABE8 in these assays.

Titers and pooled selection agree on the ranking. $n = 3$ independent transformations; bars: mean $\pm$ s.d.

"Surpasses seven rounds" is a comparison to an evolved reference, not a matched campaign-cost trial.

Breadth does not guarantee every task improves

Enzyme annotations

At 80% masking, published SI Table S4 reports joint class/fold-confidence success of 62% versus 55% for the ten common EC classes, and 72% versus 58% for random classes (guided versus ESM3). For the least represented classes, success is 33% versus 36%: guidance cannot be assumed to overcome sparse property information.

CATH fold annotations

The structure-token experiment extends guidance beyond amino-acid sequences. Correct fold classification is evaluated separately from designability (self-consistency r.m.s.d. $\leq 2\,\text{Å}$). A desired annotation alone does not establish that a backbone is realizable.

Published SI Table S4; Extended Data Figs. 1–2 and their captions. No new wet-lab enzyme/fold validation.

Which claims are secure, and which remain open?

1. **Method:** a useful unification supports modular conditioning. Exactness assumes the specified sampling process and accurate conditional predictors; TAG adds approximation.
2. **Computation:** oracle gains and diversity are encouraging. Out-of-distribution designs can exploit predictor errors; fold confidence does not validate function.
3. **Experiment:** one feedback cycle yields improved bacterial selection activity. Generality across genomic loci, sequence contexts, mammalian cells and off-target profiles remains open.

Published SI Fig. S3 calibrates rounds using 109 shared controls ($R^2 = 0.55$); bootstrap uncertainty matters for library-wide comparisons. Individual validation is a particularly valuable control.

Published SI Fig. S3; Fig. 5f; interpretation of the demonstrated assay scope.

Journal club discussion and source boundary

Discussion

Would guidance still win when predictor uncertainty is included? Which next experiment would distinguish a stronger editor from improved survival in this selection assay? How should we budget diversity versus property gain?

Reading these notes

Journal citation verified: published 29 July 2026. The journal body was subscription-gated. Method explanations use the full 85-page arXiv v4; results use the published figures/captions and 24-page published supplement. Journal and preprint baselines differ, particularly for

stability and PbrR. Published panels take precedence here; unverified journal-only method changes remain an access gap.

Journal: doi.org/10.1038/s41587-026-03207-z; preprint: arxiv.org/abs/2505.04823v4; source URLs and QA in README.md.